

Earnings Surprise, Firm Valuation, and Market Efficiency: Evidence from China

Corporate Finance Team Project

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FIN803 Corporate Finance

- ▶ Earnings guidance updates expected future cash flows and should affect firm value.
- ▶ In semi-strong efficiency, public guidance information should be priced quickly.
- ▶ China A-share is a useful setting for testing whether pricing is immediate or delayed.
- ▶ The goal is not to maximize t-stats, but to estimate a defensible valuation-adjustment effect.

Core Question

Do cleaner guidance-surprise signals predict post-announcement abnormal return in China A-shares?

- ▶ Research question is unchanged.
- ▶ What changed is the empirical design: we now prefer a cleaner signal and a shorter event window.
- ▶ Final framing emphasizes limited short-run valuation adjustment, not strong long-run PEAD.

- ▶ Firm value reflects discounted expected future cash flows.
- ▶ Guidance matters because it updates expected profitability.
- ▶ Prices should react to the surprise component relative to expectations, not to raw earnings news alone.

- ▶ Under semi-strong efficiency, public information should be incorporated quickly.
- ▶ If abnormal return appears shortly after announcement, that suggests incomplete immediate adjustment.
- ▶ But a long-window return pattern can be contaminated by unrelated news.
- ▶ So the identification strategy matters for how strong the inference can be.

- ▶ Sample period: 2020-01-01 to 2026-04-21.
- ▶ Test run shown here: 300 stocks, 5,162 raw guidance rows, 2,858 candidate events.
- ▶ Event types:
 - guidance_initial
 - guidance_upward_revision
- ▶ Event trading date: first trading day after announcement.

- ▶ Exclude ST / *ST firms.
- ▶ Require at least 120 trading days since listing.
- ▶ Require tradable event-day return and non-zero event-day volume.
- ▶ Apply 20-day turnover-based liquidity filter.
- ▶ Purpose: reduce microstructure noise and make inference more credible.

Why the Baseline Was Noisy

- ▶ Old signal `ES_main` used a noisy proxy for expectations.
- ▶ Old headline outcome `CAR60` absorbed too much non-event information.
- ▶ Pooled long-window regressions were easy to over-interpret as strong PEAD.
- ▶ Group comparisons like “moderate beats extreme” were not stable enough to be the main conclusion.

Why the Improved Specification Is More Defensible

- ▶ ES_std better standardizes guidance surprise intensity.
- ▶ CAR5 focuses on short-run valuation adjustment close to the announcement.
- ▶ Industry FE and year-quarter FE control for common shocks.
- ▶ Firm-clustered SE improve inference when firms appear repeatedly.
- ▶ This design is better suited to a cautious academic interpretation.

Preferred Variables

- ▶ Abnormal return: $AR_{i,t} = R_{i,t} - R_{m,t}$
- ▶ Preferred event window: $CAR5 = \sum_{t=1}^5 AR_{i,t}$
- ▶ Preferred signal: `ES_std`
- ▶ Controls: log size, beta

Preferred Interpretation

We test whether cleaner standardized surprise predicts short-run abnormal return, not whether large surprise mechanically implies strong long-run drift.

$$CAR5_i = \alpha + \beta_1 ES_std_i + \beta_2 \log(size_i) + \beta_3 beta_i + \text{Industry FE} + \text{YearQuarter FE} + \varepsilon_i$$

- ▶ Standard errors clustered by firm.
- ▶ Baseline CAR60 regression is retained only as a comparison benchmark.

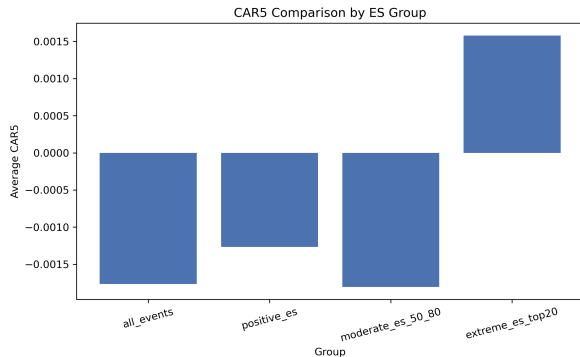
Baseline vs Preferred Specification

	Baseline	Preferred
Signal	ES_main / moderate dummy	ES_std
Outcome	CAR60	CAR5
Sample size	2,375	2,529
Key coefficient	0.0225	0.00173
p-value	0.0481	0.0195
Inference	clustered OLS	FE + clustered SE

- Preferred specification is statistically cleaner, but the economic effect is modest.

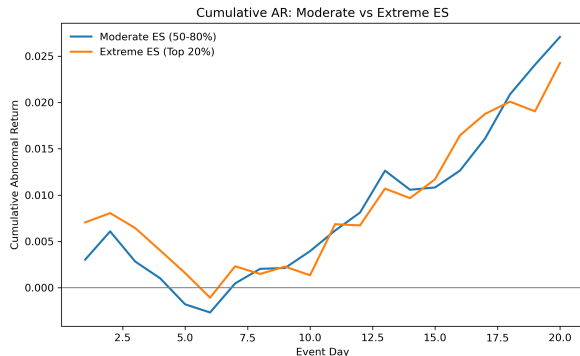
Preferred Main Result

- ▶ ES_std coefficient on CAR5: 0.001732
- ▶ p-value: 0.0195
- ▶ Sample size: 2,529
- ▶ Interpretation: cleaner standardized surprise predicts modest short-run abnormal return.



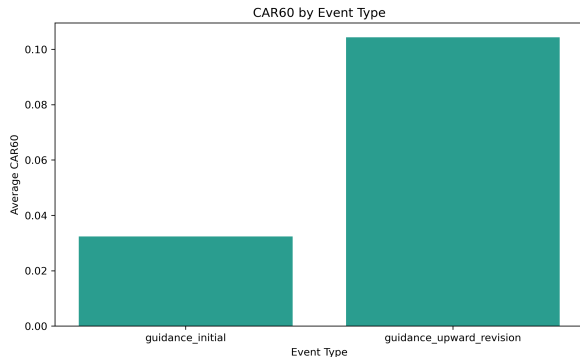
Group Evidence Is Supplementary, Not Headline

- ▶ Improved group means at CAR5 are small.
- ▶ Moderate group CAR5: about -0.18%
- ▶ Extreme group CAR5: about 0.16%
- ▶ So monotonic or dummy-group claims should not be the main narrative.



Baseline as Robustness / Cautionary Comparison

- ▶ Baseline CAR60 result remains positive and borderline significant.
- ▶ But long-window evidence is much easier to contaminate with later news.
- ▶ Therefore it should be treated as a robustness comparison, not the headline finding.



- ▶ Guidance surprise affects valuation because it changes expectations about future profitability.
- ▶ The preferred evidence supports limited short-run underreaction, not a dramatic inefficiency story.
- ▶ Better signal construction matters as much as better regression design.
- ▶ The result is statistically meaningful, but economically modest.

- ▶ Main result: cleaner standardized guidance surprise predicts modest but statistically significant short-run abnormal return.
- ▶ Best framing: limited short-run valuation adjustment after cleaner guidance surprise.
- ▶ Not supported as a headline claim: strong long-run PEAD, strong market inefficiency, or stable monotonic group ordering.
- ▶ Next step: improve the expectation benchmark further with better analyst-consensus data.